Date: ……………

Practical No: 1: Execute Basic TCP/IP utilities and commands. (eg: ping, ipconfig, tracert, arp, tcpdump, whois, host, netstat, nslookup, ftp, telnet etc...)

A. Objective:

To utilize basic TCP/IP utilities and commands to troubleshoot network issues, gather configuration information, and analyze connectivity, aiding in effective network management and diagnostics.

B. Expected Program Outcomes (POs): PO1, PO2, PO5, PO8, PO11

C. Expected Skills to be developed based on competency:

The practical is expected to develop the following skills:

- **Network Troubleshooting**: Identify and fix connectivity issues using tools like ping and arp.
- **IP Configuratio**n: View and manage IP settings with ipconfig (Windows) or ifconfig (Linux).
- **Routing Analysis**: Trace packet routes using tracert (Windows) or traceroute (Linux).
- **ARP Management**: Understand and troubleshoot ARP for address resolution.
- **Traffic Analysis**: Capture and analyze network traffic with tcpdump.
- **Domain Information**: Retrieve domain and IP details using whois, host, and nslookup.
- **Connection Monitoring**: Monitor network connections and ports using netstat.
- **File Transfer**: Transfer files using FTP and understand basic telnet for remote connections.

D. Expected Course Outcomes (Cos): CO1

E. Practical Outcome (PRo)

Execute Basic TCP/IP utilities and commands.

F. Expected Affective domain Outcome (ADos)

1. Follow coding standards.
2. Demonstrate working as a leader/ a team member.
3. Follow ethical practices.

G. Prerequisite

Theory: TCP/IP

Model:

Explanation: The TCP/IP model is a conceptual framework that standardizes the functions of a telecommunication or computing system into different layers. It consists of the Application, Transport, Network, Data Link, and Physical layers.

Example: The process of sending an email involves multiple layers of the TCP/IP model, from the application layer (email client) to the physical layer (network hardware).